ME599 W25: LiDAR Mount Redesign

December 2nd, 2025

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Agenda

1. Background
2. Engineering Methodology
3. Integration
4. Recommendations
5. Lessons Learned

Background

UWAFT

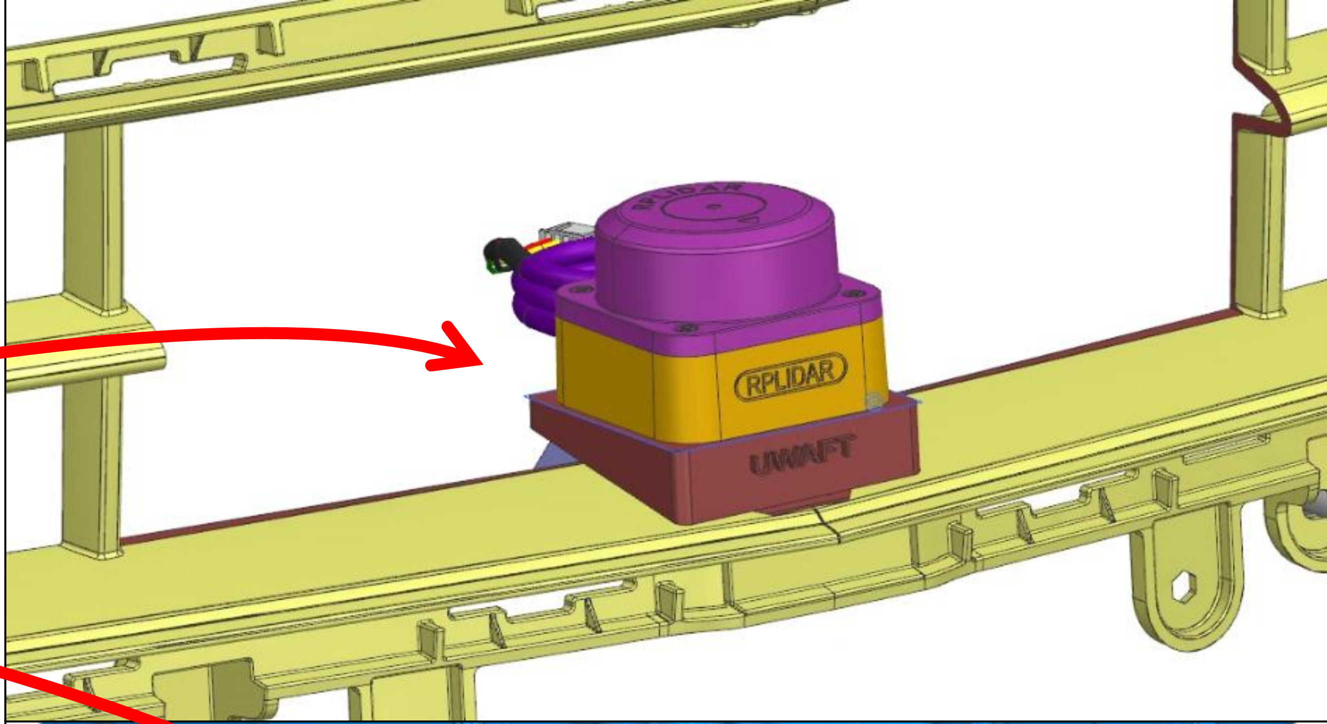
Background

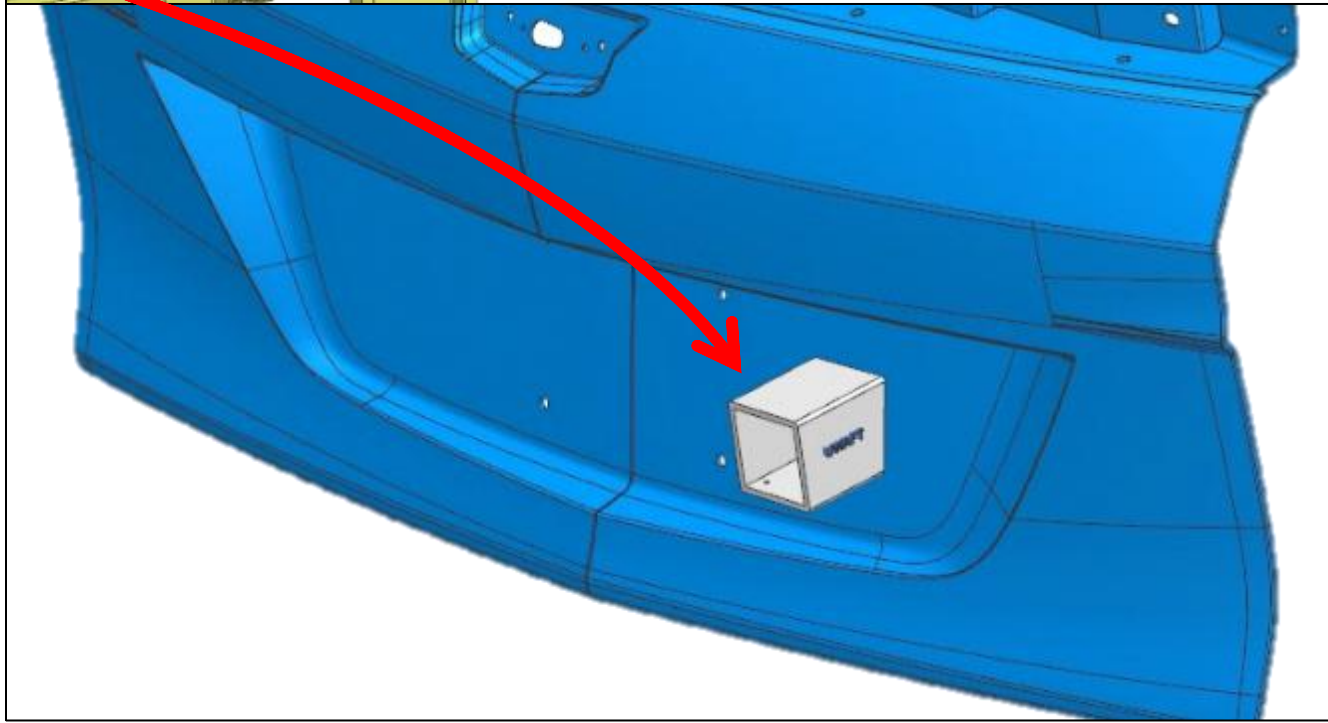
- The UWAFT team wants to mount two LiDAR sensors on the front and back
- This would enable CAV to use the LiDAR for low-speed parking scenarios
- UWAFT goals is to implement these two LiDAR sensors with keeping functionality and aesthetics in mind

Problem Definition

Existing front/rear mount issues:

- Poor accessibility
- Blocked field of view
- Target LiDAR position not met
- Not aesthetically pleasing
- Installation Issues





Design Constraints

1) Mount Angle ≤ 10°

2) Field of View ≥ 120°

3) Modular Design

4) Must fit in 3” Envelope

Design Criteria

1) Weatherproofing

2) Aesthetics

3) Structural Rigidity

4) Cost Effectiveness

Timeline

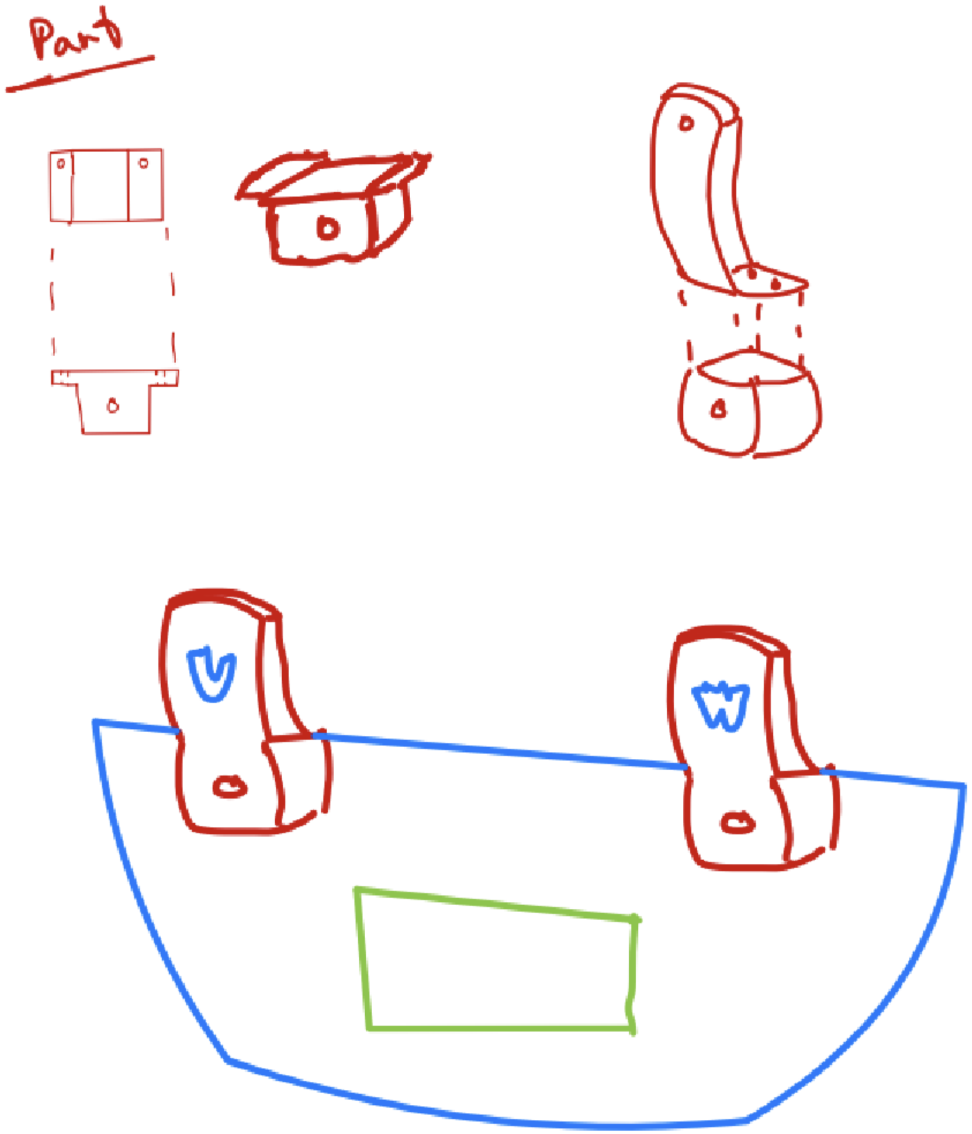
Task ID	Task Title	Start Date	Due Date	Pct pf Task Complete	September				October				November				December			
					W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16
1	**LiDAR Placement**		2025-10-06																	
1.1	Positional Research / Background Research	09-22-25	09-26-25	100%																
1.2	Define Design Requirements	09-22-25	09-26-25	100%																
1.3	Location Mapping / Prelim CAD Dimensions	10-01-25	10-06-25	100%																
2.0	Prototyping																			
2.1	Design																			
2.2	Software Learning (CAD, Siemens NX)	09-25-25	10-28-25	100%																
2.3	Mount Design - Front	10-06-25	10-28-25	100%																
2.4	Mount Design - Rear	10-06-25	10-28-25	100%																
2.7	Simulations																			
2.8	Vibrations	10-29-25	11-03-25	100%																
2.9	FEA	10-30-25	11-04-25	100%																
2.11	Vehicle Setup For Mounting																			
2.12	Protoyping	11-05-25	11-12-25	100%																
2.13	Integration																			
2.14	Electrical Routing	11-05-25	11-12-25	0%																
3	Aestheics																			
3.1	Material Selection	10-27-25	11-03-25	100%																
3.2	Surface Modeling	11-03-25	11-10-25	100%																
3.3	Manufacturing Methods	11-10-25	11-17-25	100%																
4.0	Testing and Calibration																			
4.1	LiDAR Calibration/ Validation	11-17-25	11-24-25	0%																

Engineering Methodology

Design Ideation

Concept Sketches

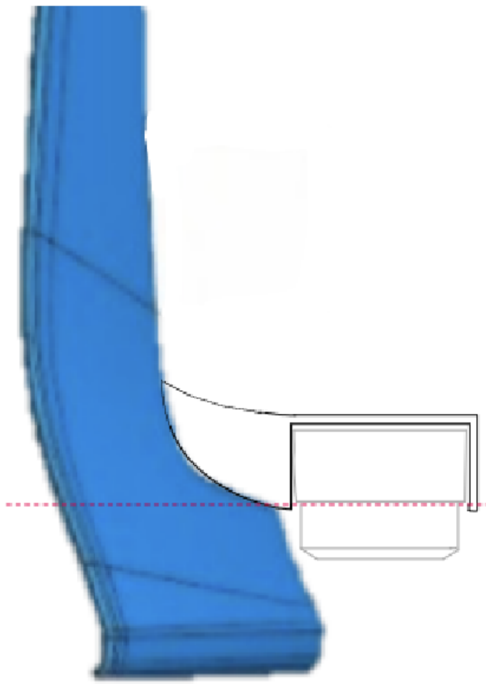
Sketch #1



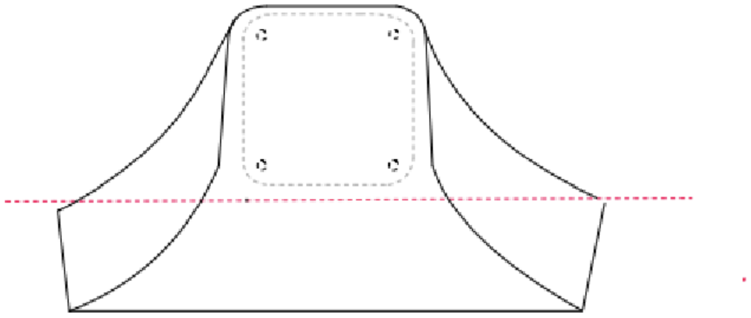
- under the trunk handle
- focus on curvature
- lidar → plate → hook mount
- focus on material selection

Sketch #2

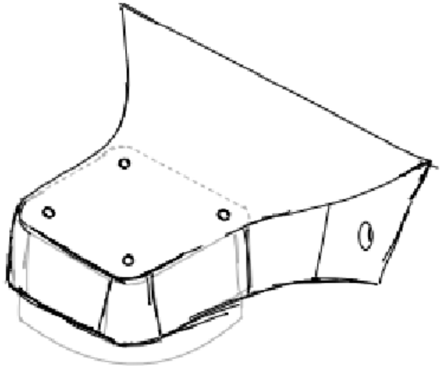
Side View



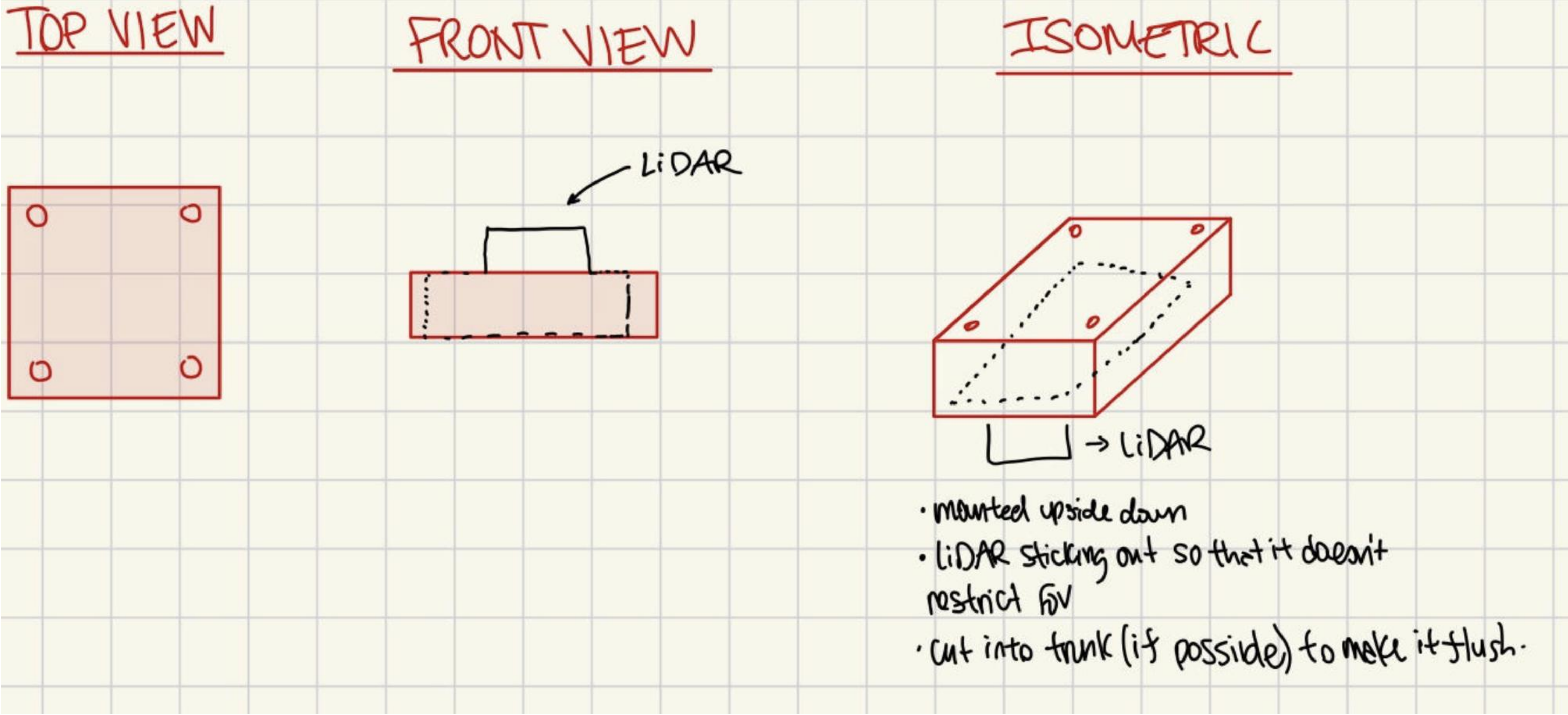
Top View



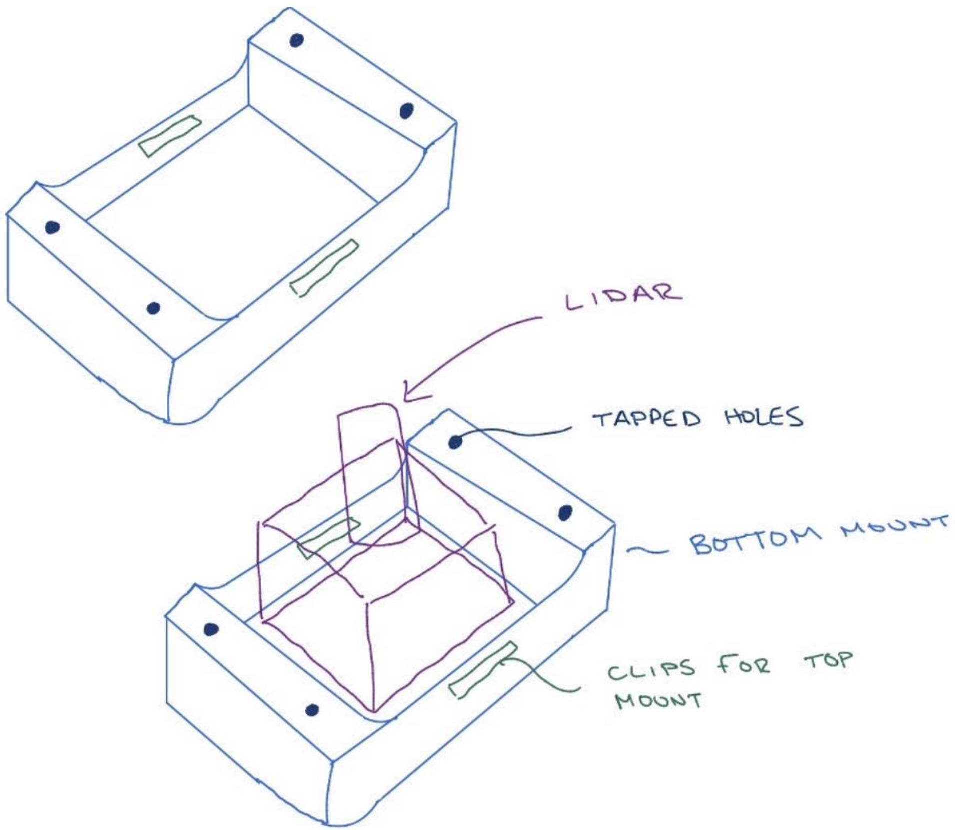
Isometric View

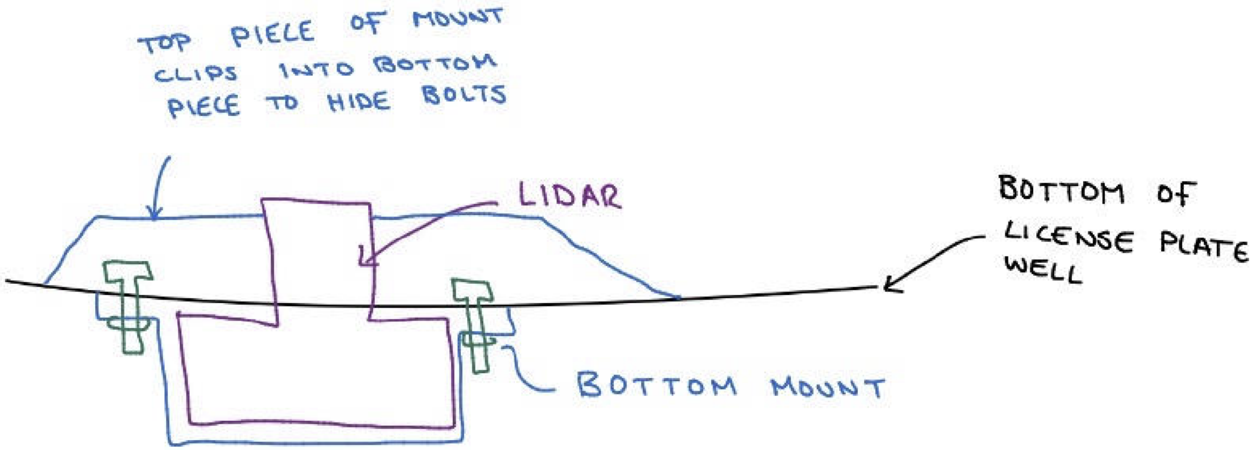


Sketch #3


TOP VIEW
FRONT VIEW
LiDAR
ISOMETRIC
→ LiDAR
· mounted upside down
· LiDAR sticking out so that it doesn't restrict FOV
· cut into trunk (if possible) to make it flush.

Sketch #4


LIDAR
TAPPED HOLES
BOTTOM MOUNT
CLIPS FOR TOP
MOUNT


TOP PIECE OF MOUNT
CLIPS INTO BOTTOM
PIECE TO HIDE BOLTS
LIDAR
BOTTOM OF
LICENSE PLATE
WELL
BOTTOM MOUNT

Decision Matrix

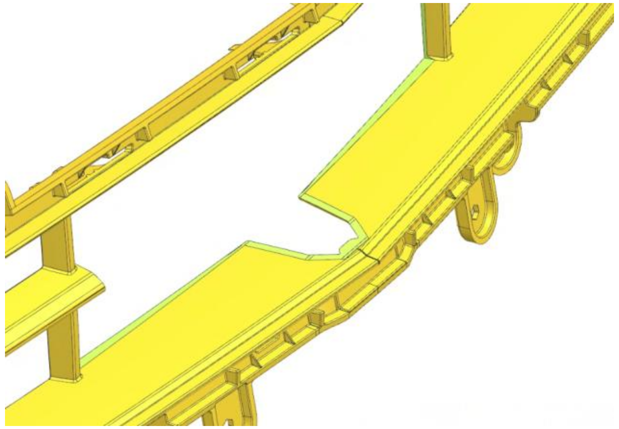
Design Criteria	Sketch #1	Sketch #2	Sketch #3	Sketch #4
Manufacturability	2	3	1	3
Durability	2	3	2	3
Aesthetics and integration with existing vehicle	3	3	1	4
Sensing and FOV	3	4	3	2
Serviceability and ease of access	2	3	2	1
Total Score	12	16	9	13

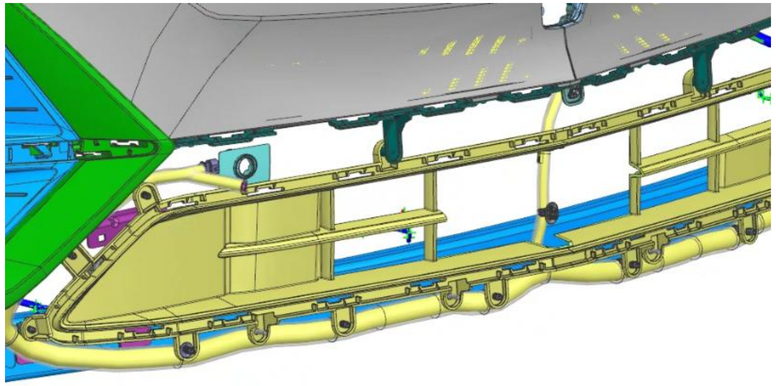
Design Modifications
Front & Rear Mount

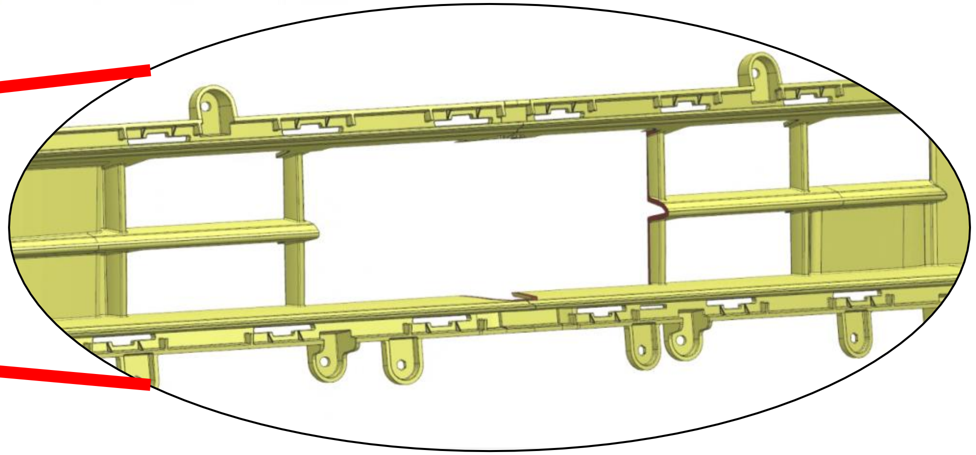
Front Mount Design

Front Mounting Location: Inside the lower fascia air inlet

(center plastic section is removed for LiDAR clearance)

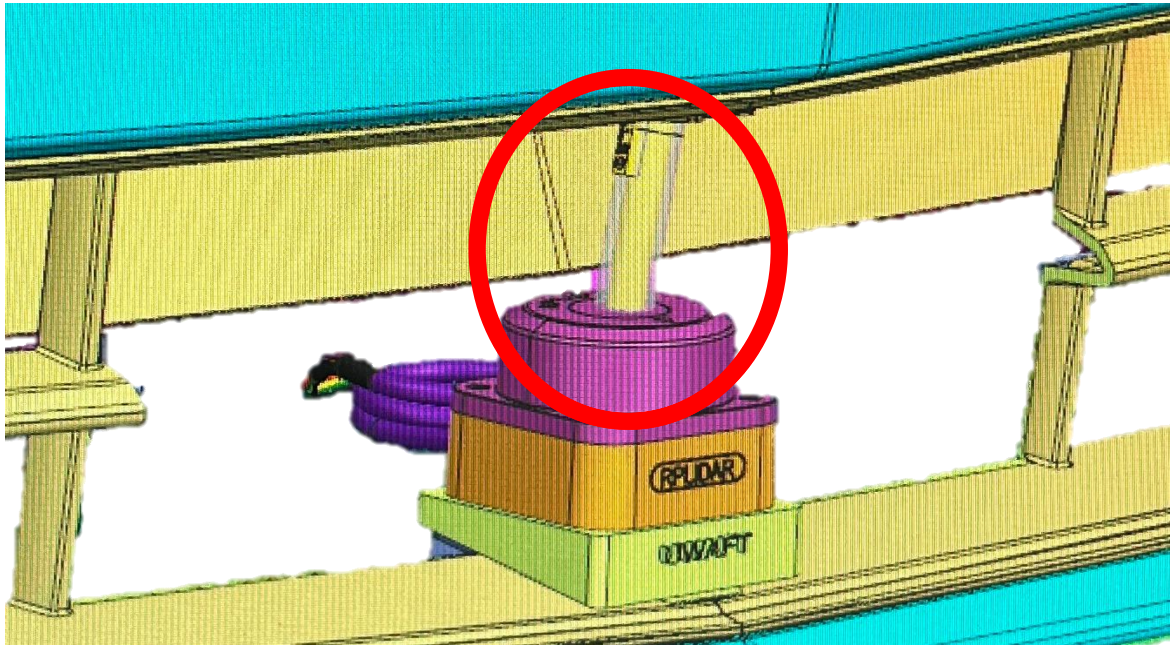


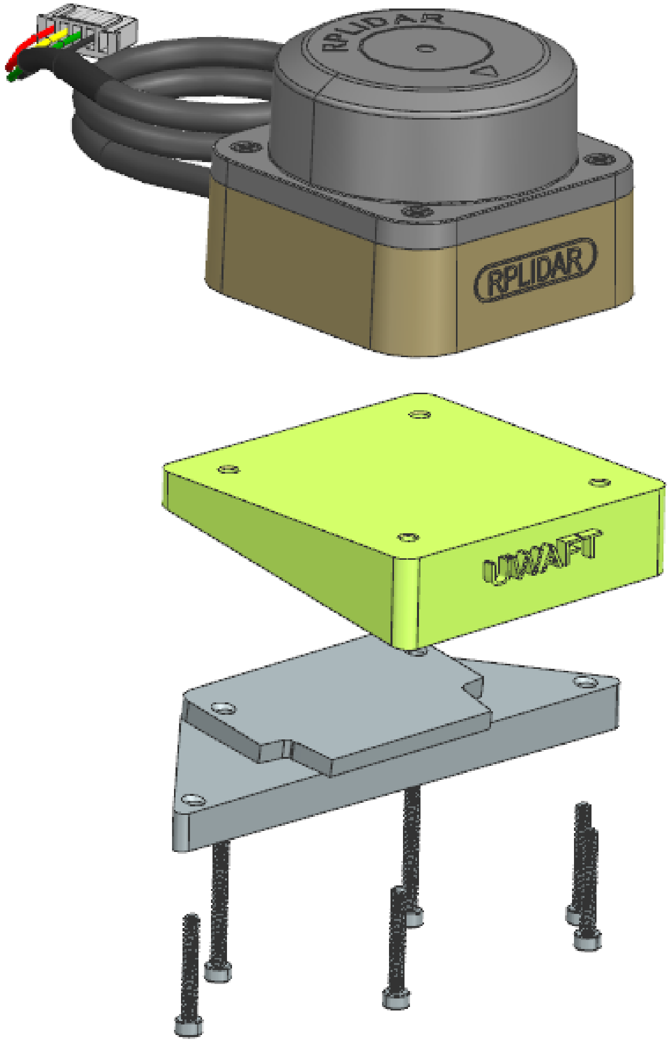




Obstacles & Challenges

Front Mount Installation – Old Design

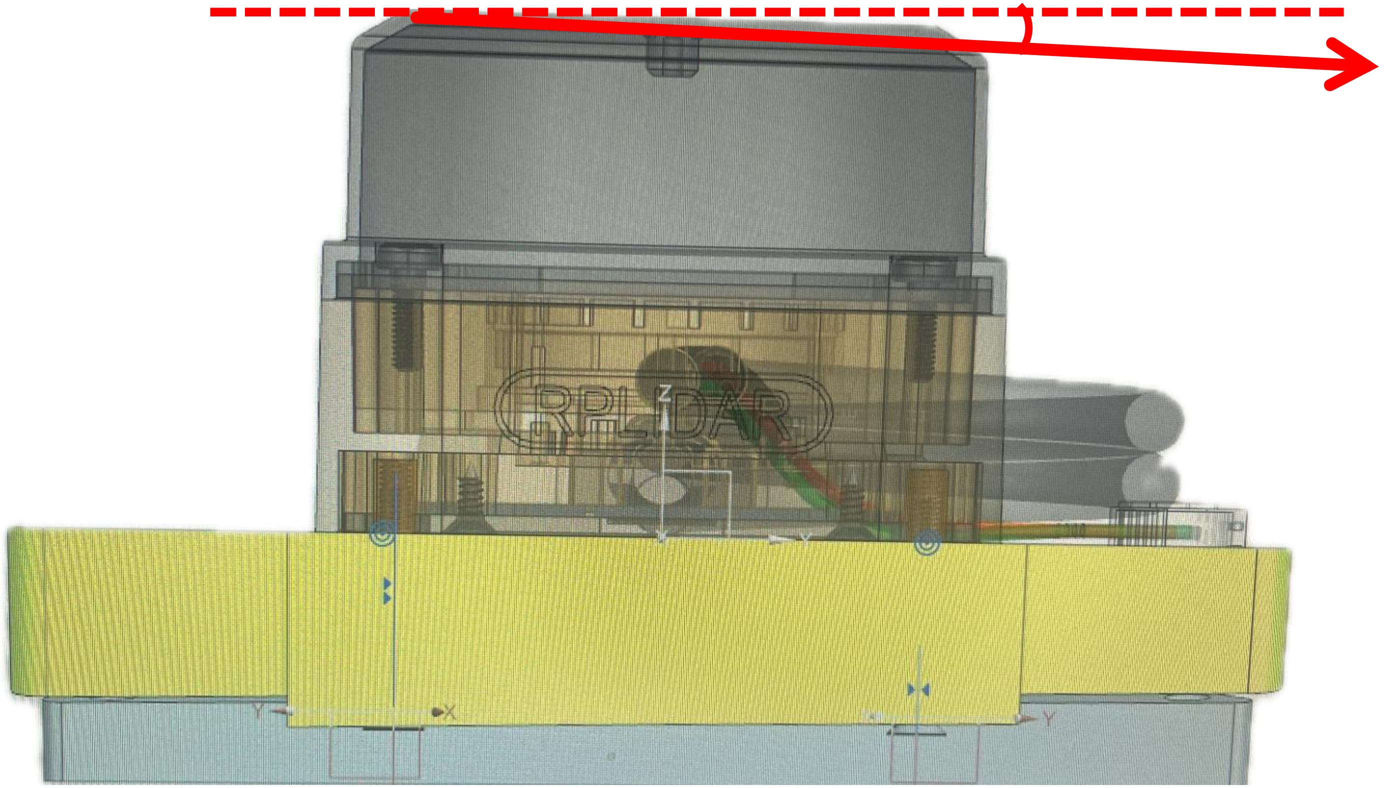




- Team did not want to remove front bumper
- It was impossible to install the previous teams design without removal of bumper
- Previous design did not account for the cable system currently installed

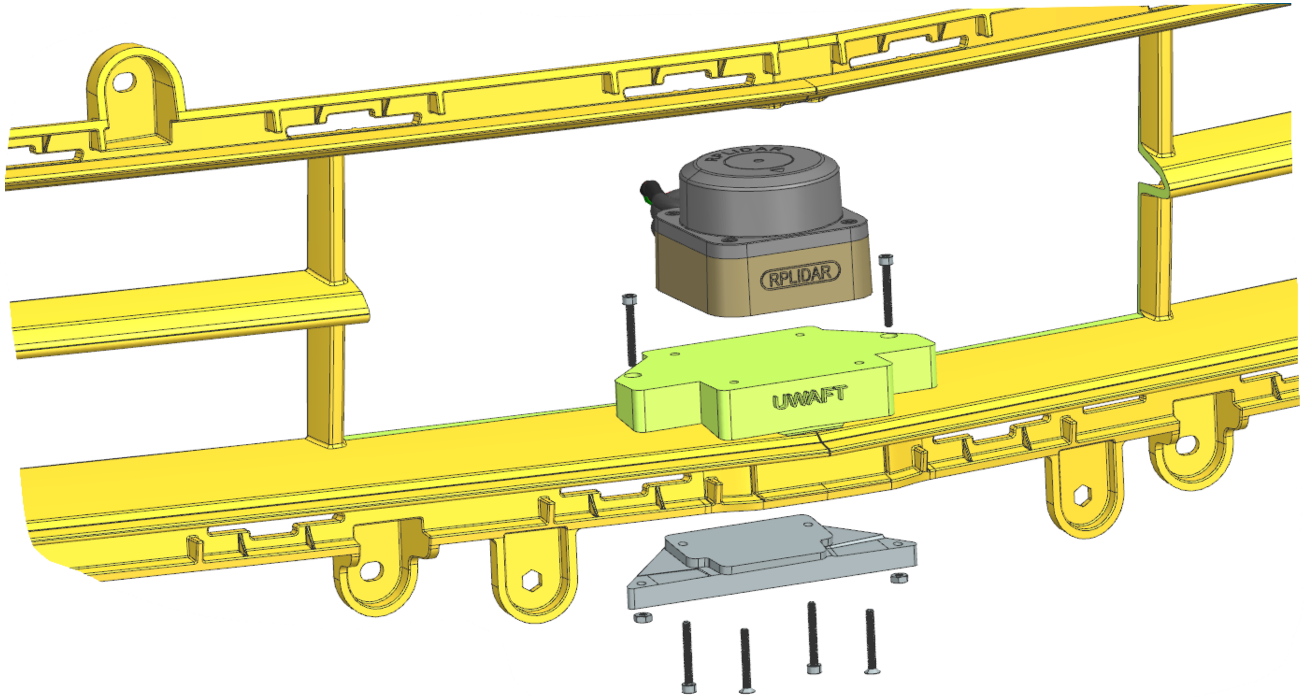
Obstacles & Challenges

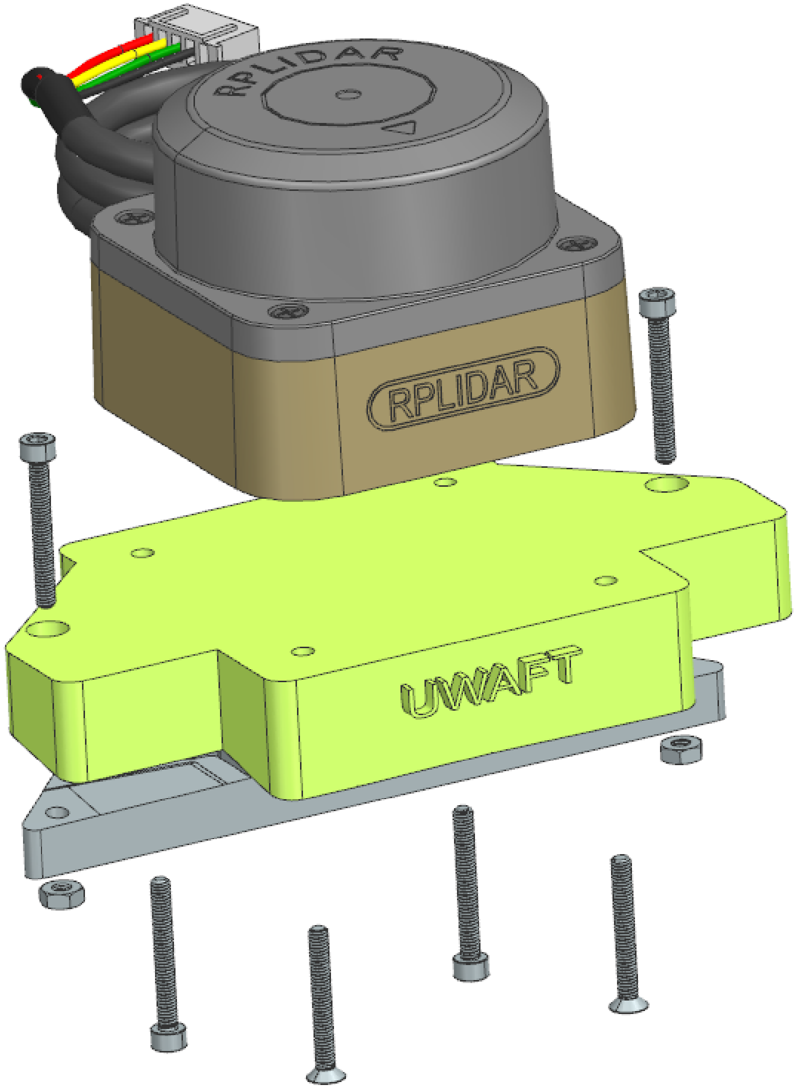
Front Mount – Old Design



- Previous design had geometry flaws → height of mount was not symmetrical

New Front Mount CAD

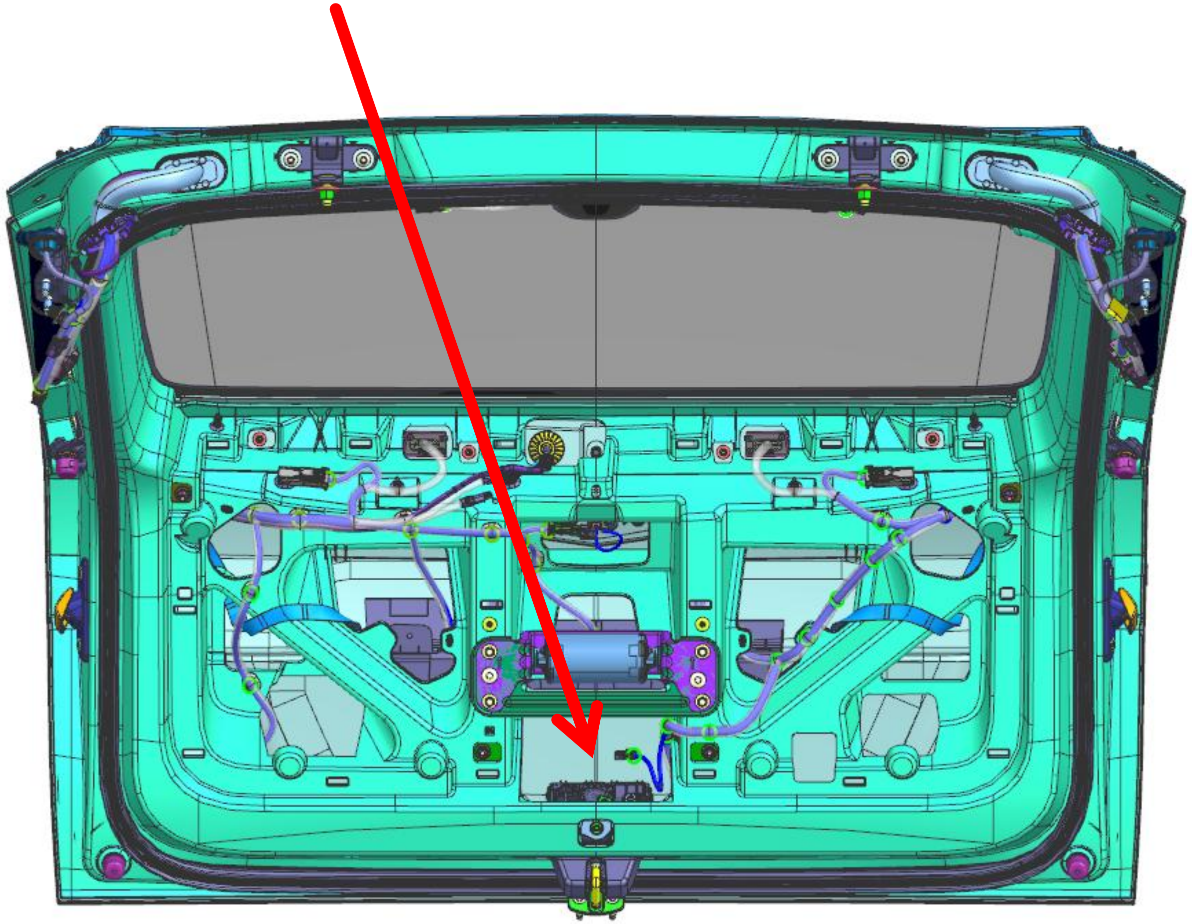




- LiDAR is mounted parallel to the ground plane
- Mount screws are accessible

Rear Mount Design

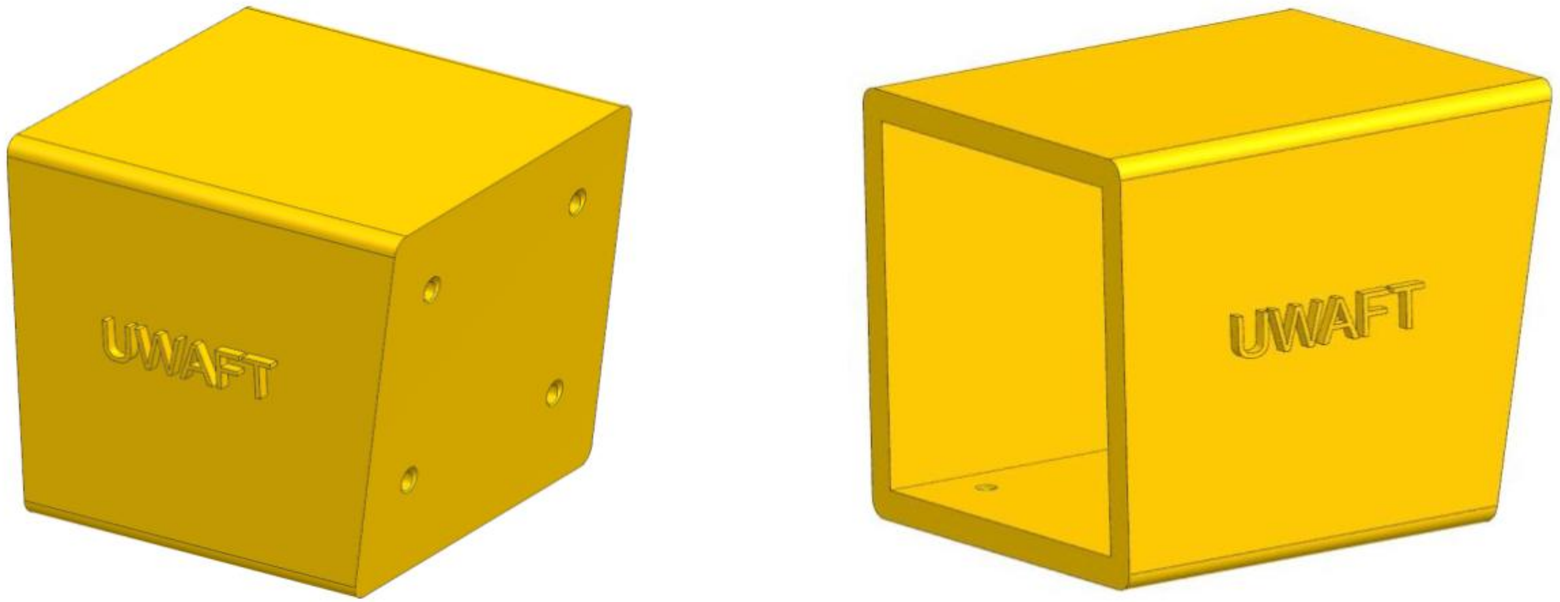
Rear Mounting Design: Underneath the license plate bolting into the trunk



Obstacles & Challenges

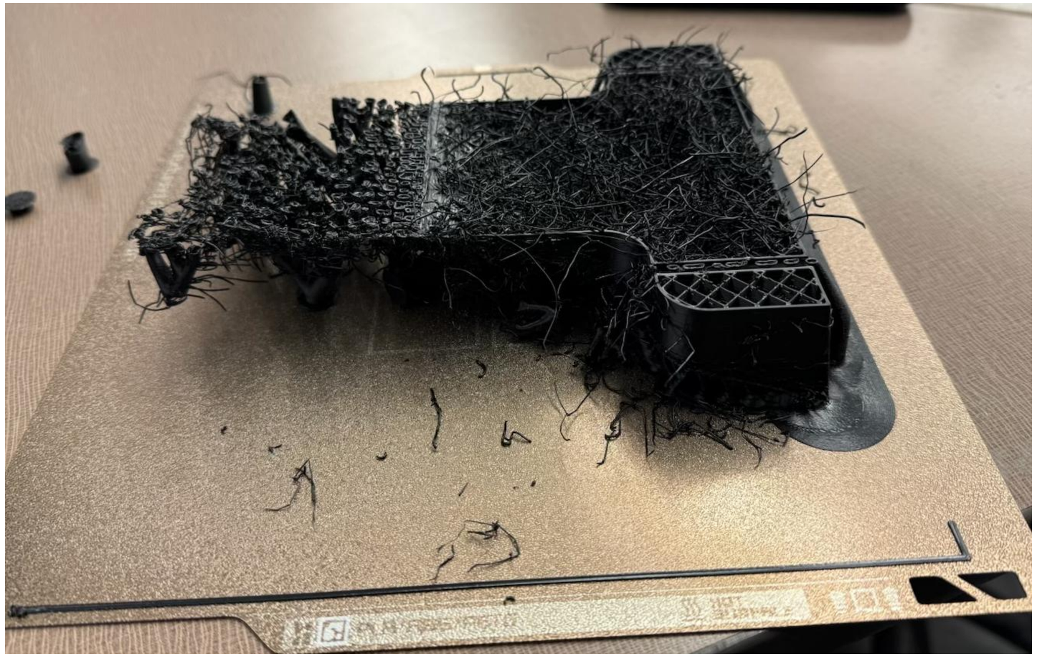
Rear Mount Design – Old Design

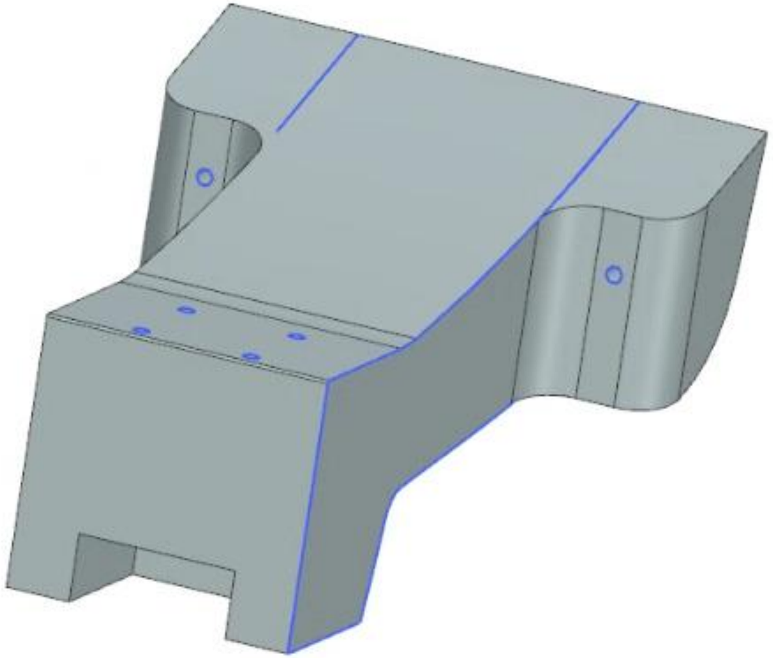
- Was **not** aesthetically pleasing
- **Limited FOV**



Obstacles & Challenges

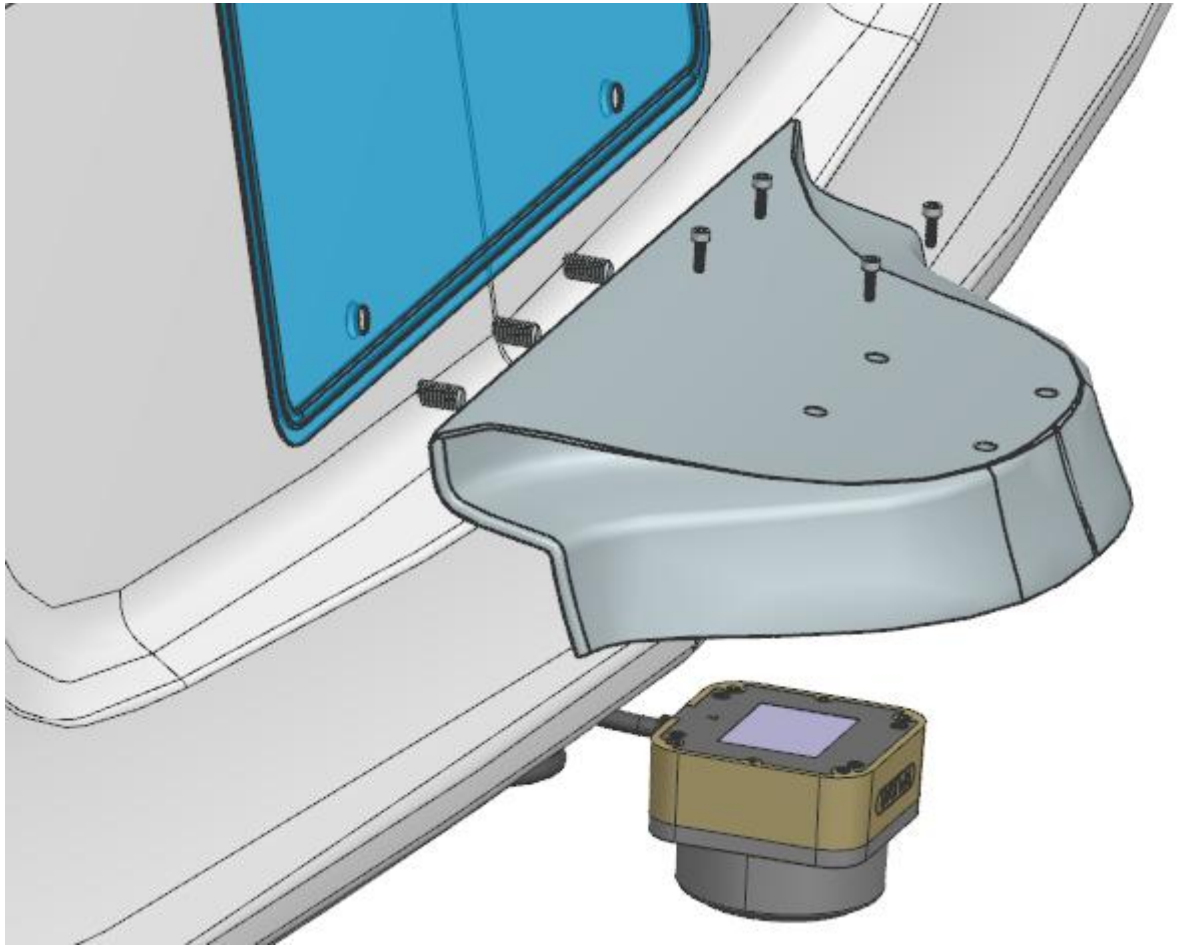
1st Revision: Rear Mount Design

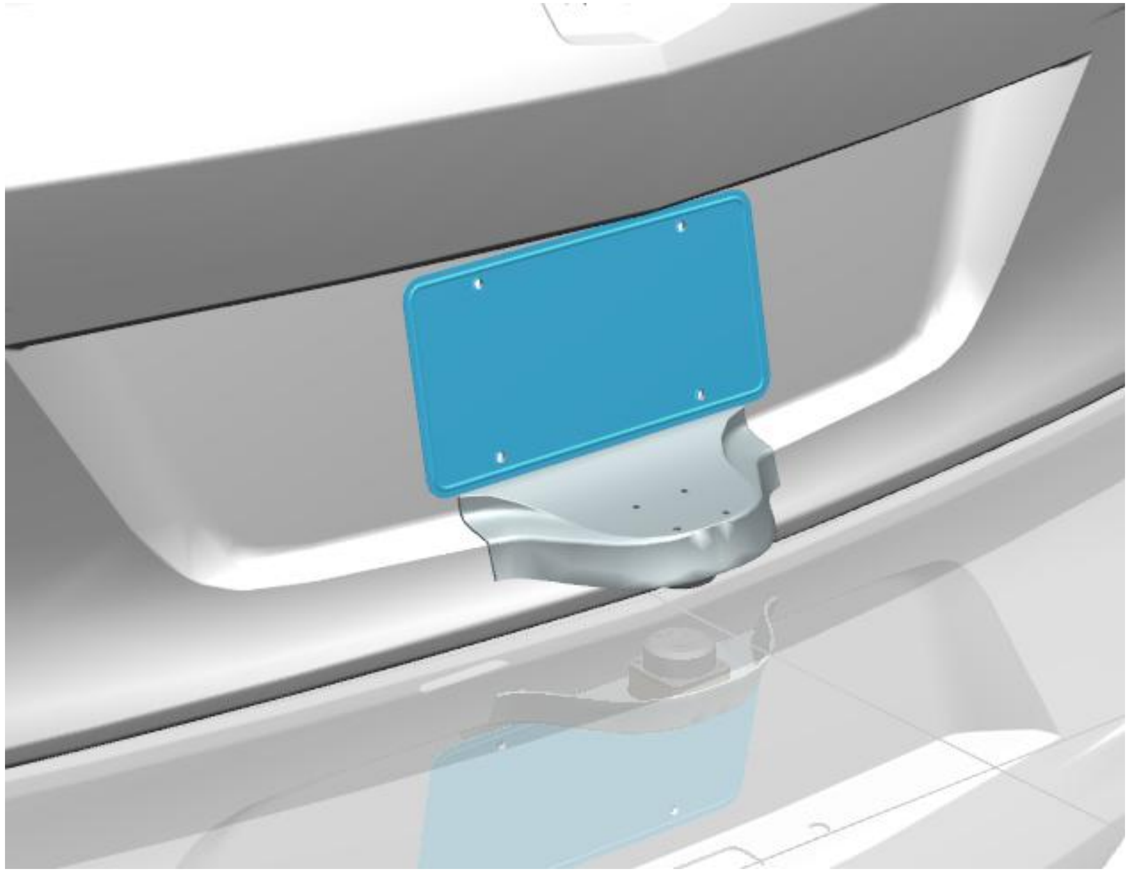




- First Iteration of Rear Mount Design failed printing multiple times
- Difficult to print and print time was over 25+ hours

New Rear Mount CAD





ANSYS Simulations

Design Validation

Design Validation

Simulation Parameters

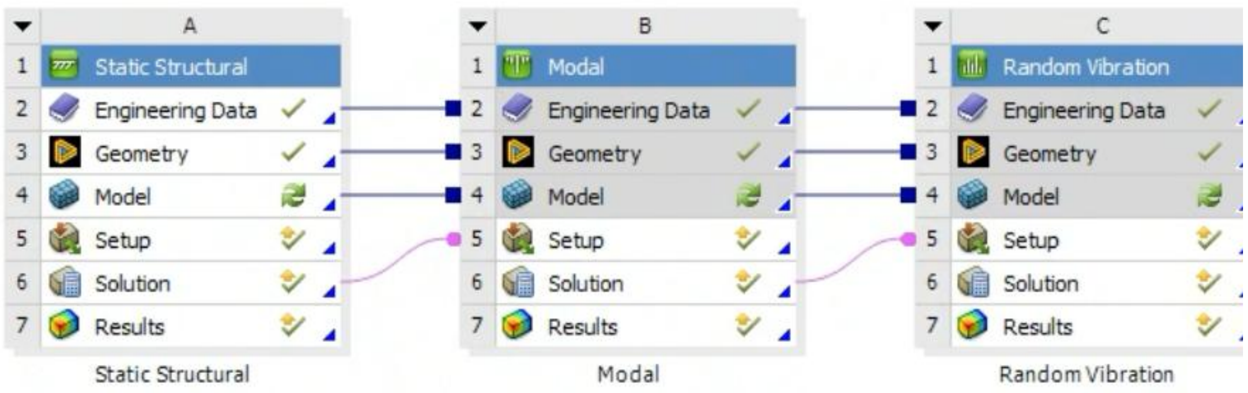


Figure 12: Example for Cross-Axis Acceleration Upper Limit for Random Vibration - Sprung Mass

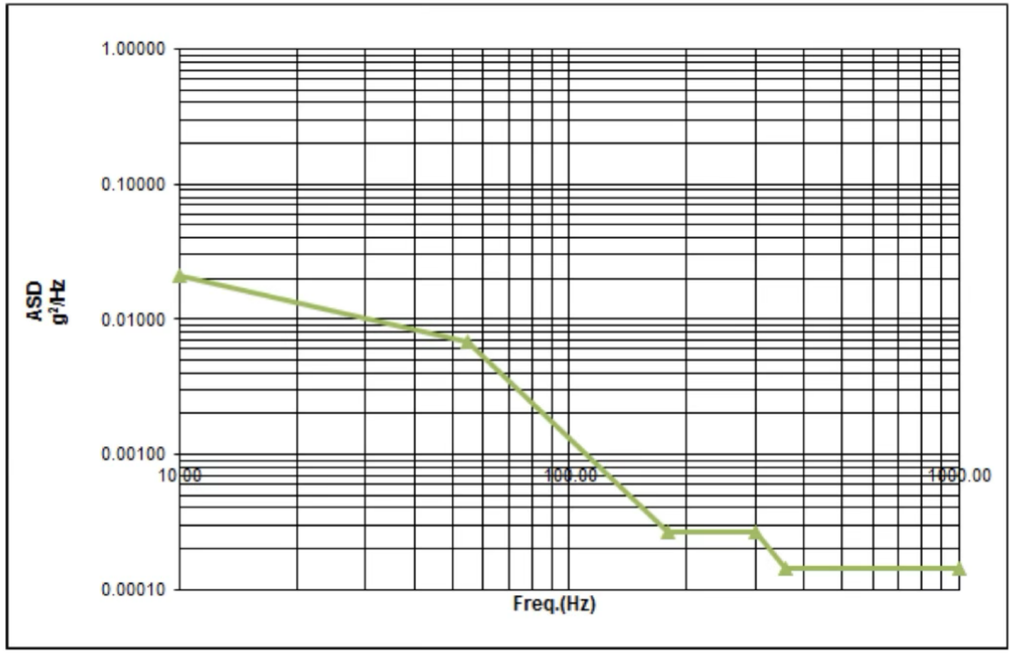


Table 18: Random Vibration Cross-Axis Acceleration Upper Limit for Sprung Mass

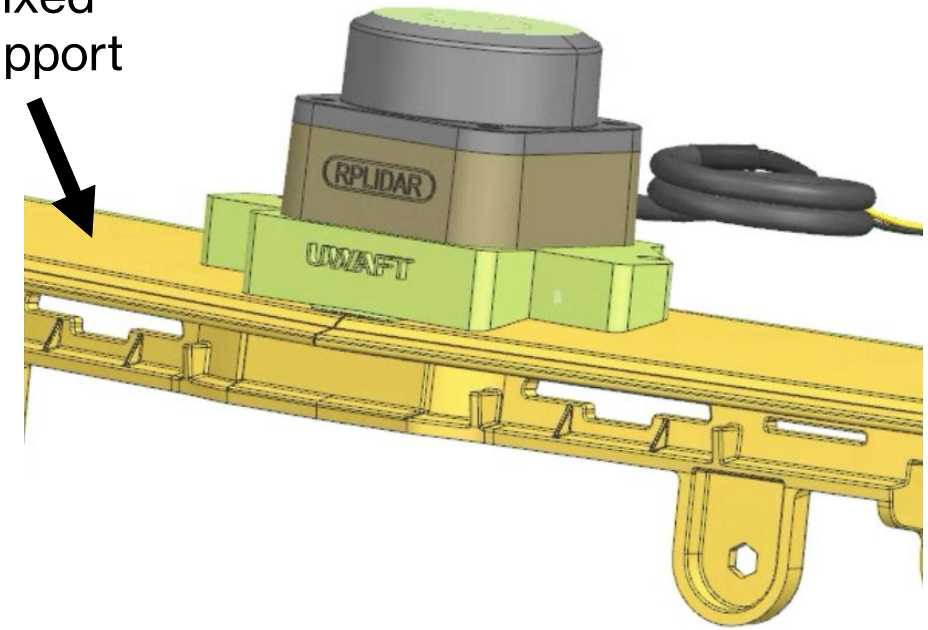
FREQ in Hz	PSD in $(m/s^2)^2/Hz$	PSD in g^2/Hz
10	2.0098	0.0209
55	0.6543	0.0068
180	0.0253	0.0003
300	0.0253	0.0003
360	0.0136	0.0001
1000	0.0136	0.0001

Effective Acceleration = 8.8 m/s^2 = 0.9 G_{RMS}

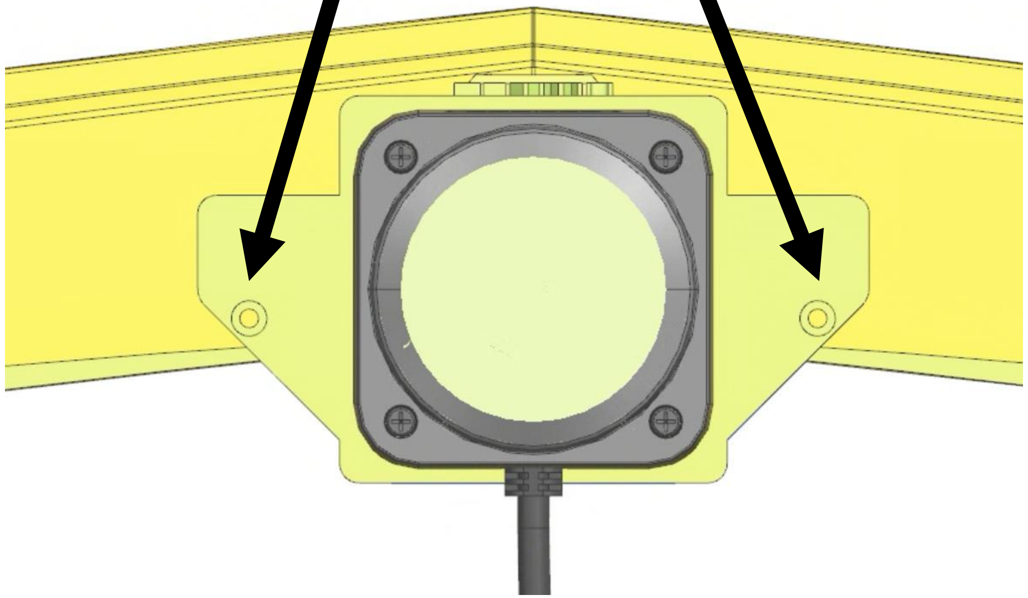
Front Mount Design Validation

Model Setup

Fixed
Support

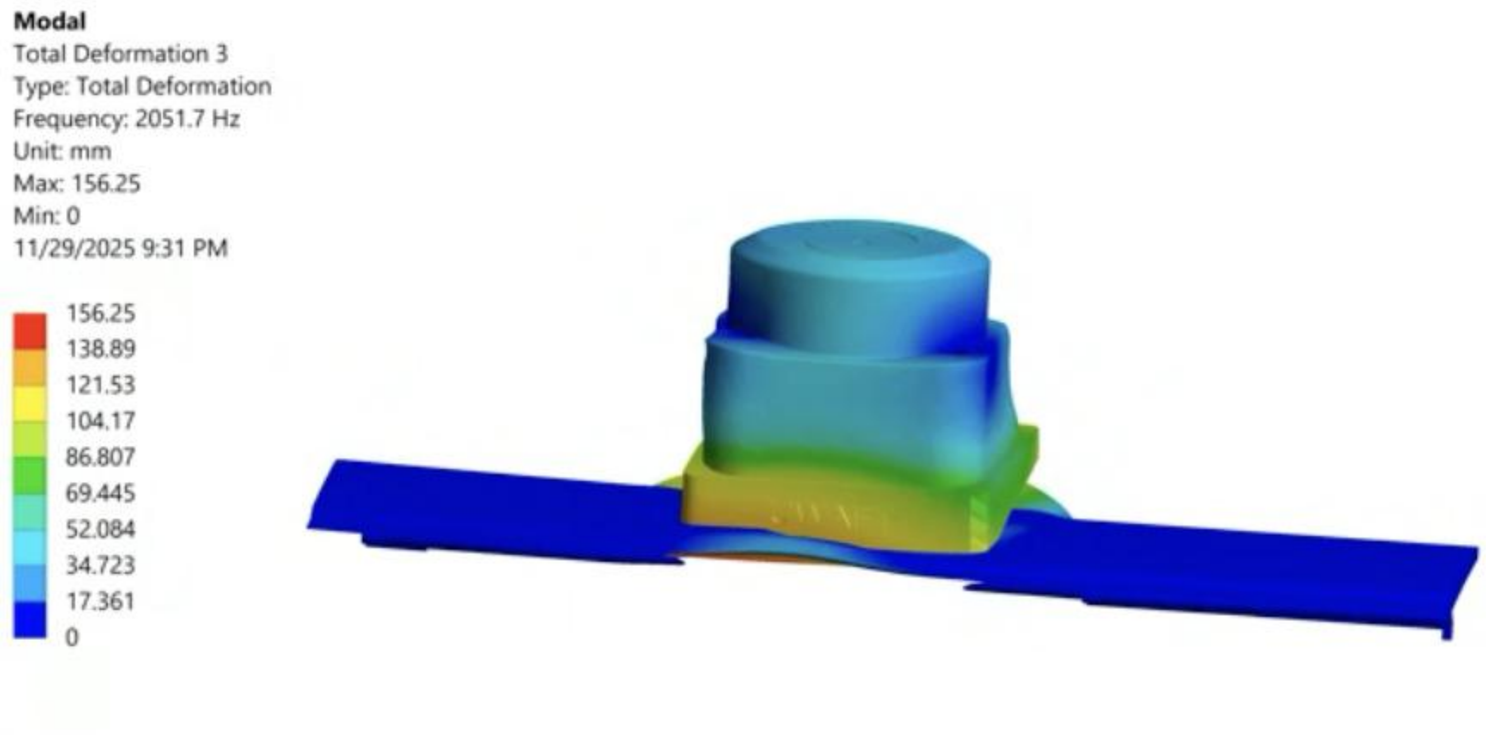


Bolted Joint
Connection
(Preload Applied)



Front Mount Design Validation

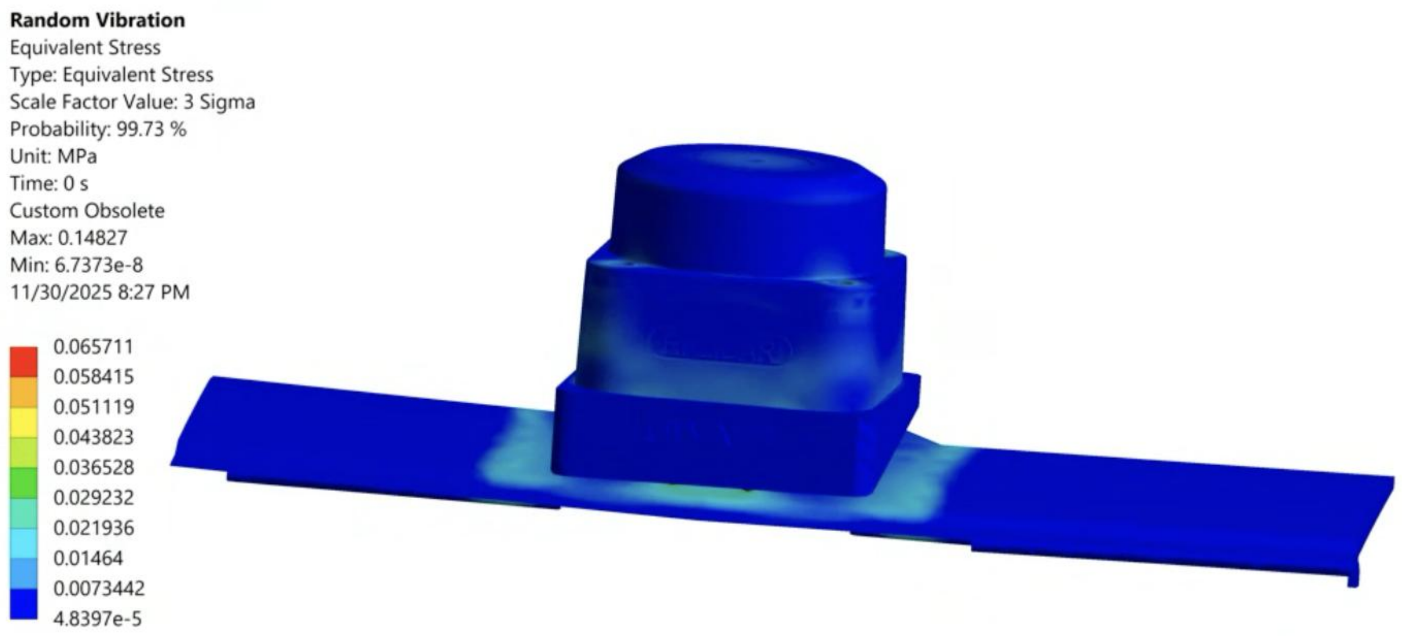
Modal Results

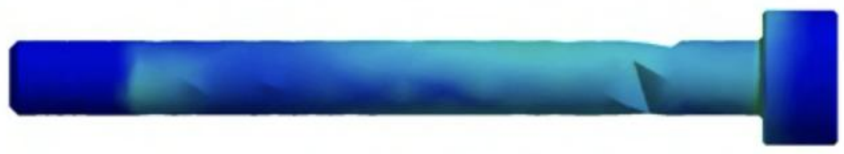


	Mode	Frequency [Hz]
1	1.	1305.4
2	2.	1566.3
3	3.	2051.7
4	4.	2436.8
5	5.	2630.
6	6.	2799.2

Front Mount Design Validation

Random Vibration: Stresses

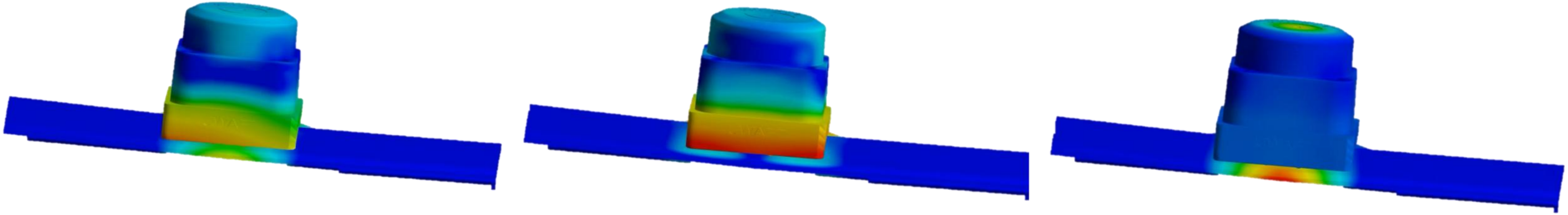




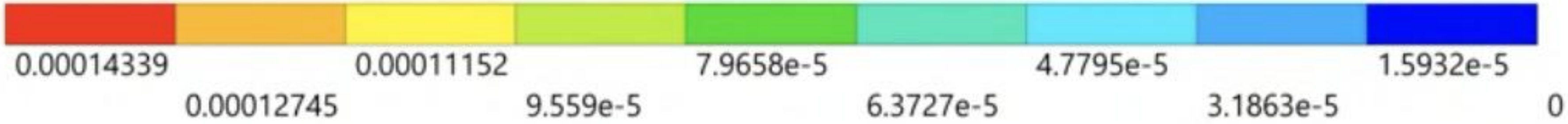
Bolt mounting to Bumper

Front Mount Design Validation

Random Vibration: Directional Deformation

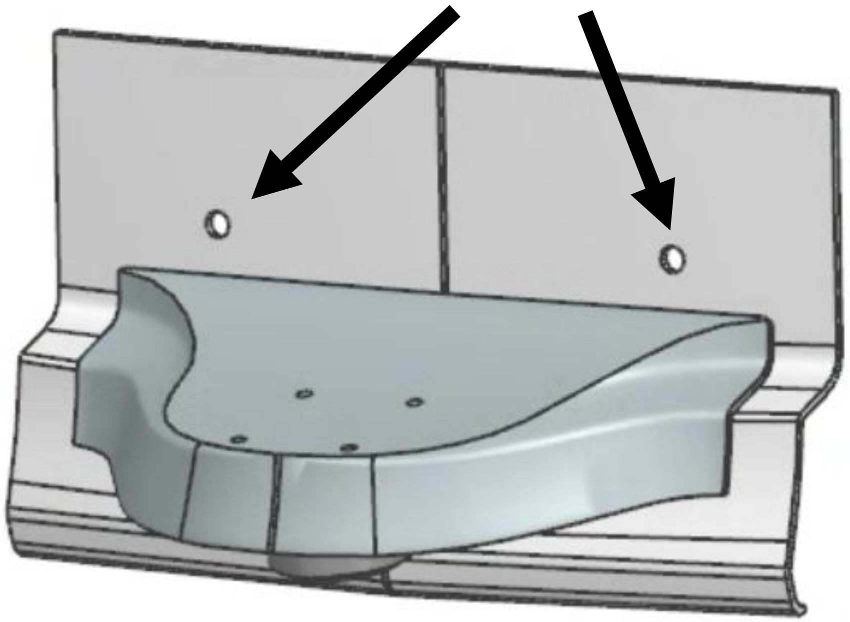


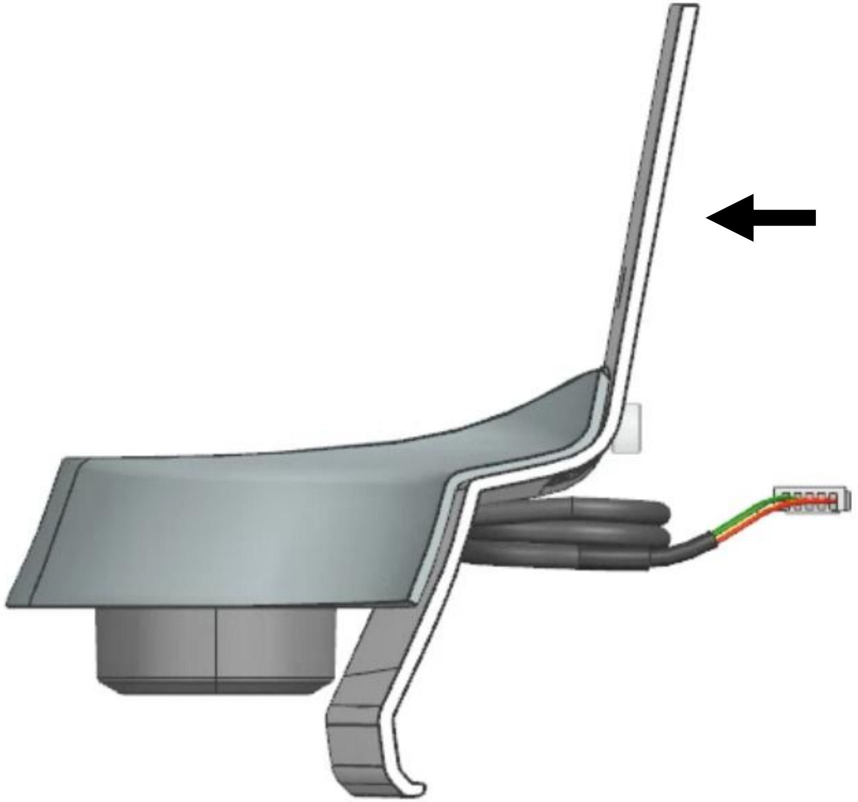
X-Axis **Y-Axis** **Z-Axis**



Rear Mount Design Validation

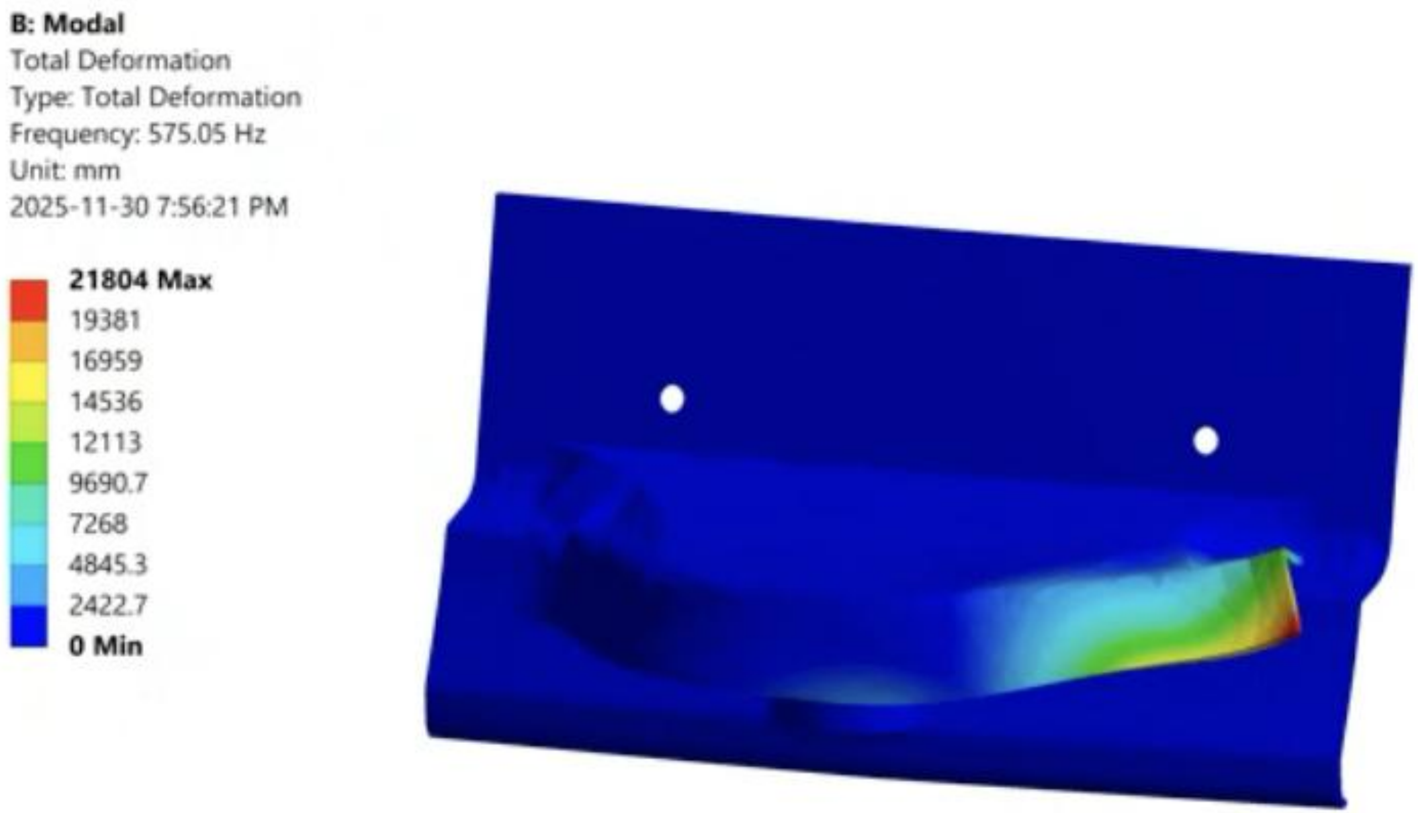
Model Setup





Rear Mount Design Validation

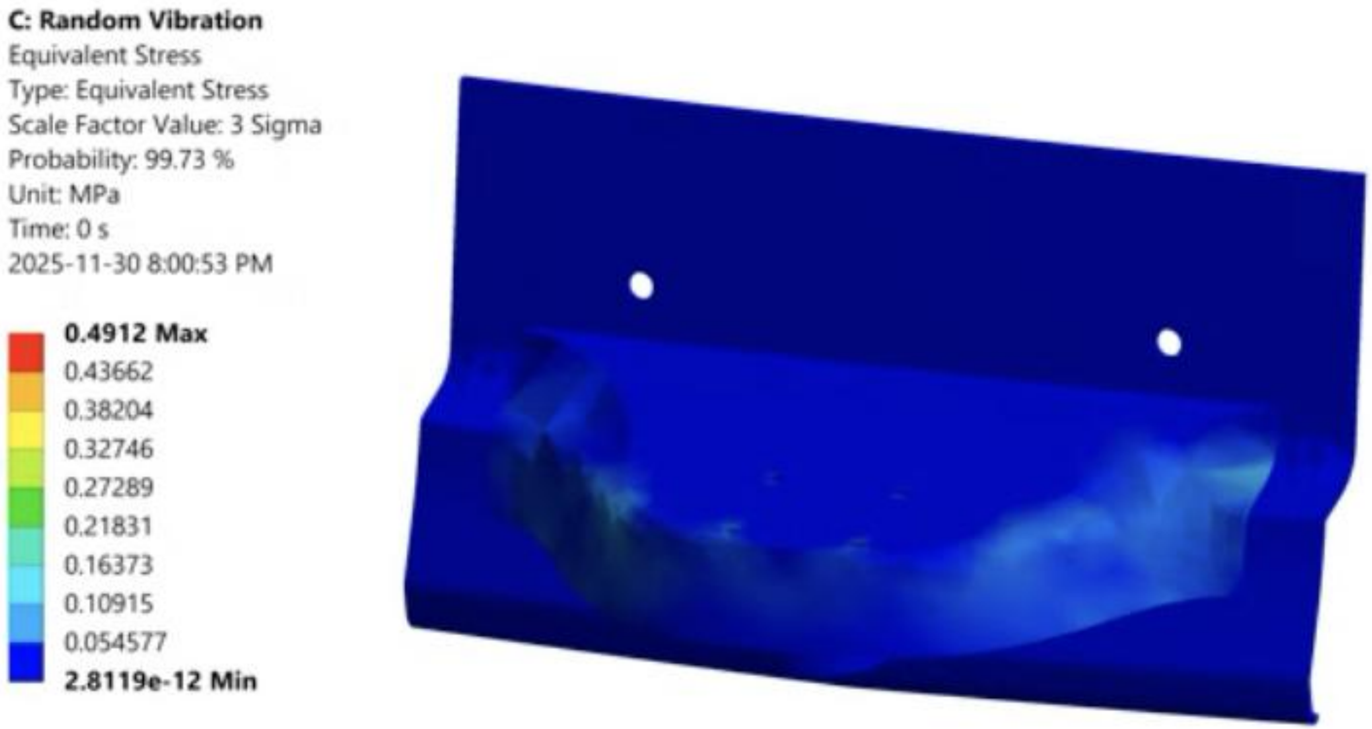
Modal Results



	Mode	Frequency [Hz]
1	1.	575.05
2	2.	656.88
3	3.	805.56
4	4.	959.17
5	5.	997.73
6	6.	1071.9

Rear Mount Design Validation

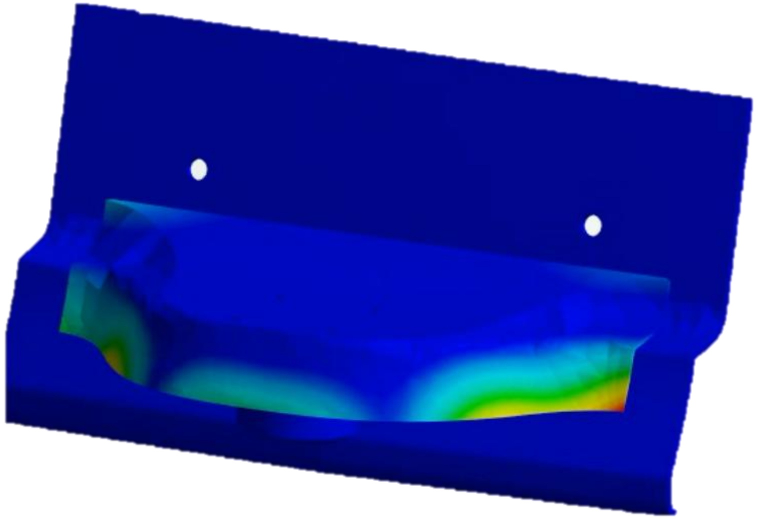
Random Vibration: Stresses



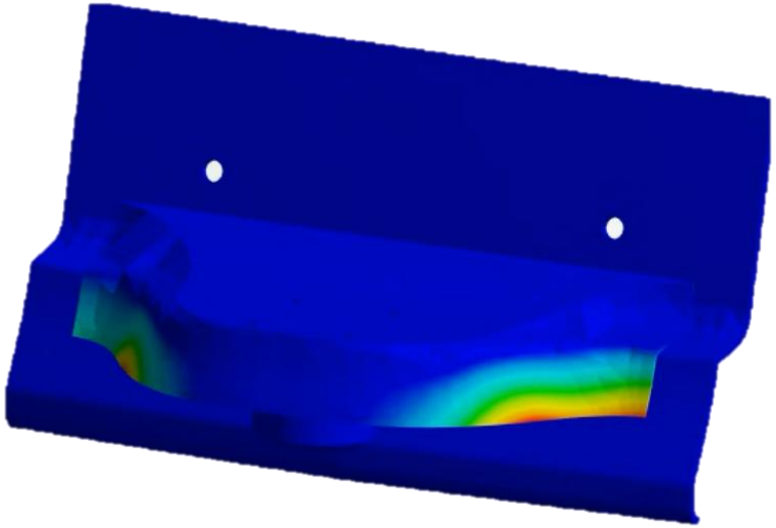
Bolt mounting to vehicle

Rear Mount Design Validation

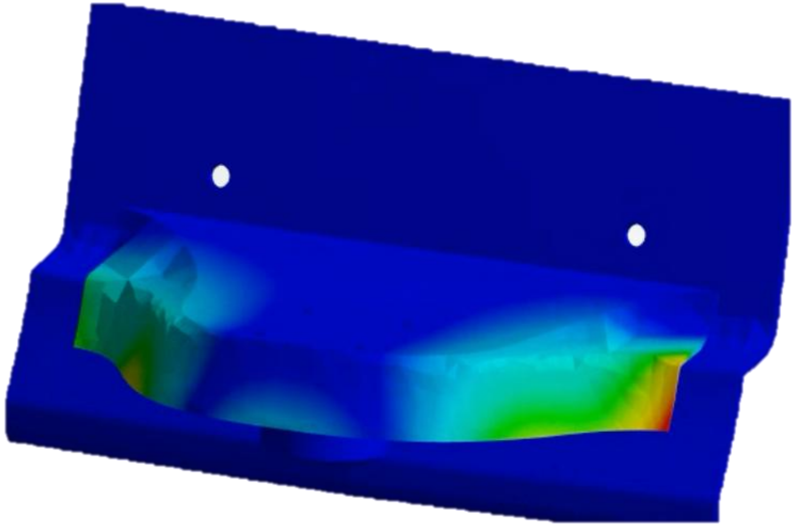
Random Vibration: Directional Deformation



X-Axis



Y-Axis

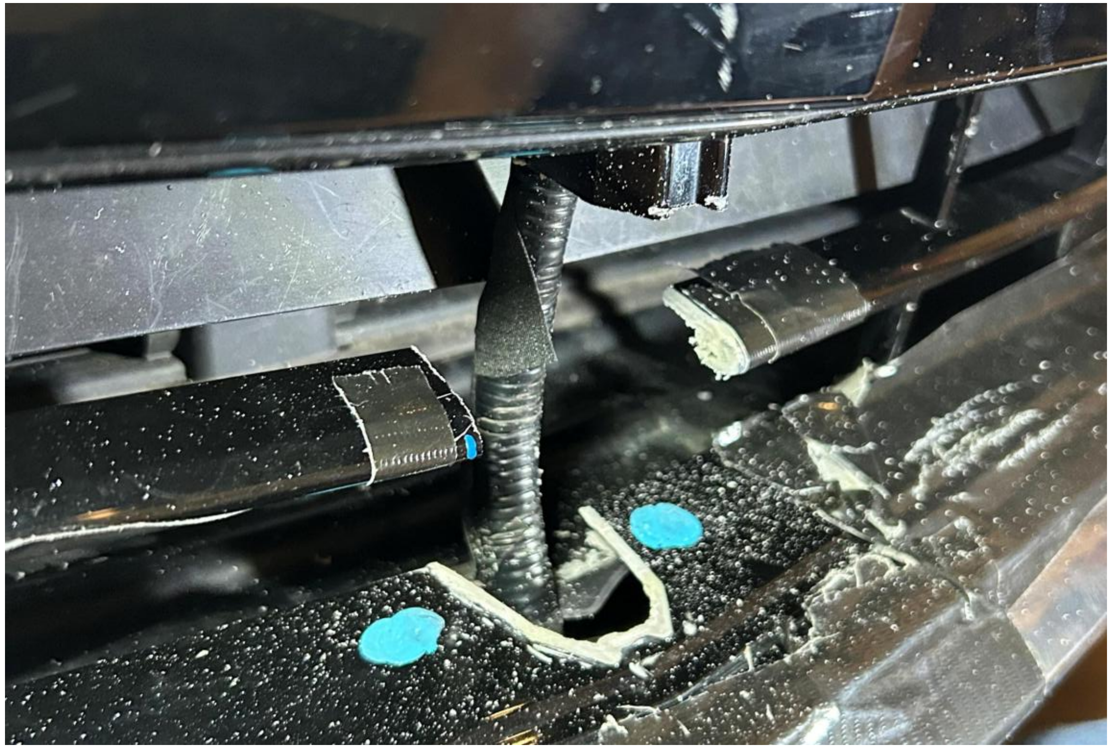


Z-Axis

Integration

Front Mount Fabrication

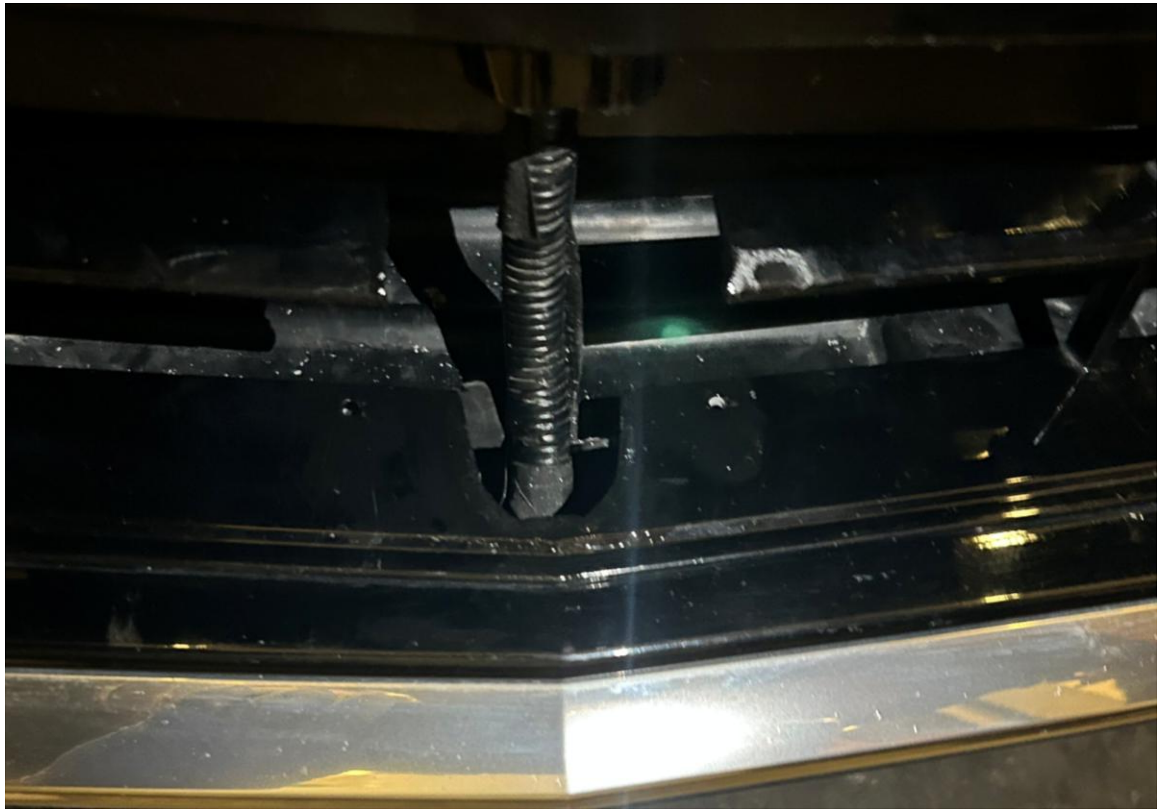
3D Printed



Cut Into Grid Mesh

Front Mount Installation

Mounted & Wire Repositioning



Repainted and Polished for Aesthetics

Front Mount Installed!!

Rear Mount Fabrication

3D Printed

Fit Testing

Recommendations

Project Recommendations

- ❑ Conduct FEA simulations to test for environmental factors
  - ❑ Thermal cycling to see effects of Thermal Expansion
  - ❑ Simulate for items hitting the mount (i.e. rocks)
- ❑ Implement all electrical components to test functionality
  - ❑ Connect and route wires through appropriate channels
  - ❑ Drill any holes required in trim pieces to cover up loose wires

Lessons Learnt

Takeaways from this Project

Lessons Learned

- Aesthetics
  - Learned about polishing and finishing plastic surfaces for aesthetics
- Project Management
  - Utilized tools such as JIRA and an Excel Tracker for organization
  - Accessibility is key, for integration

New approaches

- UWAFT Guidelines
  - There was ambiguity with project rules such as the 3” envelope, asking for help a lot earlier would have saved the team significant time
- Resource Allocations
  - Knowing about all the resources available to us and inquiring about would have helped
  - This includes 3D printing services, simulation software such as ANSYS etc.

Thank You